```
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.datasets import make_classification
from sklearn.metrics import accuracy_score
```

```
x,y=make_classification(n_samples=1000,n_features=20,random_state=42)

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)

base_learners=[]

num_base_learners=10

for i in range(num_base_learners):
  bootstarp_indices=np.random.choice(len(x_train),size=len(x_train),replace=True)
  x_bootstrap=x_train[bootstarp_indices]
  y_bootstrap=y_train[bootstarp_indices]
```

```
base_learner=RandomForestClassifier(n_estimators=10,random_state=42)
base_learner.fit(x_bootstrap,y_bootstrap)
base_learners.append(base_learner)
```

```
base_learner_predictions=[]
for base_learner in base_learners:
    y_pred=base_learner.predict(x_test)
    base_learner_predictions.append(y_pred)
```

```
ensemble_predictions=np.round(np.mean(base_learner_predictions,axis=0))

accuracy=accuracy_score(y_test,ensemble_predictions)
print("Ensemble Accuracy:",accuracy)
```

```
Ensemble Accuracy: 0.86
```

```
import numpy as np
from sklearn.ensemble import RandomForestClassifier
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```

```
df=pd.read_csv("/content/covid_toy - covid_toy.csv")
df
```

	age	gender	fever	cough	city	has_covid
0	60	Male	103.0	Mild	Kolkata	No
1	27	Male	100.0	Mild	Delhi	Yes
2	42	Male	101.0	Mild	Delhi	No
3	31	Female	98.0	Mild	Kolkata	No
4	65	Female	101.0	Mild	Mumbai	No
...	...	...	...	...	...	...
95	12	Female	104.0	Mild	Bangalore	No
96	51	Female	101.0	Strong	Kolkata	Yes
97	20	Female	101.0	Mild	Bangalore	No
98	5	Female	98.0	Strong	Mumbai	No
99	10	Female	98.0	Strong	Kolkata	Yes

100 rows × 6 columns

```
df.isnull().sum()
```

	0
age	0
gender	0
fever	10
cough	0
city	0
has_covid	0

dtype: int64

```
df['fever']=df['fever'].fillna(df['fever'].mean())
```

```
df.isnull().sum()
```

	0
age	0
gender	0
fever	0
cough	0
city	0
has_covid	0

dtype: int64

```
from sklearn.preprocessing import LabelEncoder
lb=LabelEncoder()
```

```
df['city']=lb.fit_transform(df['city'])
df['cough']=lb.fit_transform(df['cough'])
df['gender']=lb.fit_transform(df['gender'])
```

```
x=df.drop(columns=['has_covid'])
y=df['has_covid']
```

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ensemble_predictions=np.round(np.mean(base_learner_predictions,axis=0))
```

```
accuracy=accuracy_score(y_test,ensemble_predictions)
print("Ensemble Accuracy:",accuracy)
```

```
Ensemble Accuracy: 0.86
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